

HLS synthesized latency:

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=====
== Performance Estimates
=====
+ Timing:
  * Summary:
  +-----+-----+-----+-----+
  | Clock | Target | Estimated | Uncertainty |
  +-----+-----+-----+-----+
  | ap_clk | 10.00 ns | 7.300 ns | 2.70 ns |
  +-----+-----+-----+-----+

+ Latency:
  * Summary:
  +-----+-----+-----+-----+-----+-----+
  | Latency (cycles) | Latency (absolute) | Interval | Pipeline |
  | min | max | min | max | min | max | Type |
  +-----+-----+-----+-----+-----+-----+
  | 2127 | 2127 | 21.270 us | 21.270 us | 2048 | 2048 | dataflow |
  +-----+-----+-----+-----+-----+-----+
```

Co-simulation latency and clock period:

RTL	Status	Latency(Clock Cycles)			Interval(Clock Cycles)			Total Execution Time (Clock Cycles)
		min	avg	max	min	avg	max	
VHDL	NA	NA	NA	NA	NA	NA	NA	NA
Verilog	Pass	2157	2157	2157	NA	NA	NA	2157

Post-implementation resource utilization:

```
=== Post-Implementation Resource usage ===
SLICE:      0
LUT:       25887
FF:        23107
DSP:        192
BRAM:       58
URAM:       0
LATCH:      0
SRL:       1895
CLB:       6611

=== Final timing ===
CP required:      10.000
CP achieved post-synthesis: 3.808
CP achieved post-implementation: 8.190
Timing met
```

Final speedup ratio calculation:

2157 cycles * 8.190 post-imp. clock period = 0.01766583 ms

9.20587236 / 0.01766583 = 521.1117938 speedup